# Price Path Embeddings in a Hang Seng Index Portfolio

Samson Lam

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**Abstract**

This study explores whether embeddings of the recent trading history of a stock carry predictive information, and what they deliver in a portfolio. For each Hang Seng Index (HSI) stock, the daily return path is embedded in low dimensions linearly by principal component analysis (PCA), and the path with technical indicators nonlinearly by a variational autoencoder (VAE). A pooled regression maps the embedding to a forecast of next-month returns. In a long-only portfolio, the forecast selects the stocks, and the Black–Litterman model constructs the weights. Listed HSI derivatives enhance yield through delta-hedged short calls and a put spread collar held when implied volatility is high. From September 2021 to August 2026, the forecast has a rank information coefficient near 0.07, and the VAE and PCA perform similarly. The PCA portfolio with both derivatives strategies returns 6.28% a year against 3.09% for the HSI total return, an active return of 3.05% with an information ratio of 0.46. Portfolio construction alone loses 0.85% a year, stock selection adds 2.93%, and derivatives add 0.98%. Although the forecast itself is significant, its transfer into active return is not, and neither are the derivatives strategies.[1]

## 1 Introduction

Returns over the past year are known to predict returns over the following months (Jegadeesh and Titman, 1993), yet a year of daily returns and technical indicators holds far more inputs than a regression on noisy monthly returns can use. An embedding represents such a high-dimensional input in a few dimensions. It can be linear, by PCA, or nonlinear, by a VAE with long short-term memory (LSTM) layers. This study asks (i) whether embeddings of the past year of returns and technical indicators of HSI stocks carry information about their returns over the next month, (ii) whether a nonlinear embedding improves on a linear one, and (iii) what the embeddings deliver in a portfolio.

Machine learning has been widely used to forecast stock returns from past prices. Gu et al. (2020) compare linear and nonlinear methods for predicting the returns of US stocks, trace the gains of trees and neural networks to nonlinear interactions among predictors, and find momentum among the dominant signals. Jiang et al. (2023) train a convolutional network on images of past price charts and find price patterns that differ from momentum and reversal and predict returns more accurately. Fischer and Krauss (2018) train an LSTM network (Hochreiter and Schmidhuber, 1997) on the return sequences of all S&P 500 constituents to predict whether a stock will outperform the cross-sectional median on the next day, and Wu et al. (2026) feed daily features built from prices and volume to an LSTM in trend-following strategies. A VAE (Kingma and Welling, 2014) embeds such an input in a latent space of low dimension whose coordinates can serve as predictors, and PCA provides its linear counterpart.

For portfolio construction, mean-variance optimization (Markowitz, 1952) with estimated expected returns yields extreme and unstable weights. Black and Litterman (1992) propose instead to start from the expected returns implied by the market portfolio in equilibrium and to tilt them toward the investor's views in proportion to the confidence attached to each view, and He and Litterman (1999) set the uncertainty of each view proportional to the variance of the prior, which removes the need to choose the scale of the prior. This study uses the model to turn the forecasts into weights.

[1] Code and replication data are available at *GitHub*.

The evidence on option strategies for equity portfolios is mixed. Israelov and Nielsen (2015) decompose a covered call into an equity position, a short volatility position and an equity timing position, and find that the first two are compensated while the timing position is not. The short volatility position is compensated by the volatility risk premium, which Bakshi and Kapadia (2003) infer from the negative average gain of a delta-hedged long option. On protection, Israelov (2019) finds protective puts less effective at reducing drawdowns than a static reduction of exposure, and Israelov and Klein (2016) find that collars earn less equity premium than their index and have underperformed it, while Szado and Schneeweis (2010) report that a collar adjusted by momentum, volatility and economic signals outperforms a passive one. Following this evidence, the yield strategy of this study hedges the delta of its short calls, and the protection strategy holds a collar only when implied volatility is high.

This study examines a USD 100 million long-only portfolio of HSI stocks with listed HSI derivatives, benchmarked against the HSI total return (HSI TR), and decomposes its outperformance into three sources: portfolio construction by the Black–Litterman model, stock selection by the embedding forecast, and yield enhancement by derivatives strategies.

Applied to HSI stocks from September 2021 to August 2026, the study answers the three questions as follows. (i) The embeddings carry significant information, with a rank information coefficient (IC) of about 0.07 and a $t$-statistic near 2.5. (ii) The nonlinear embedding does not improve on the linear one. (iii) The PCA portfolio with derivatives strategies outperforms the HSI TR by 3.05% a year, with an information ratio (IR) of 0.46. Portfolio construction alone loses 0.85%, stock selection adds 2.93%, and the derivatives strategies add 0.98%. The forecast is significant, but its transfer into active return is not, and neither are the derivatives strategies, whose gains also depend on pricing options at the HSI Volatility Index (VHSI).

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 sets out the embeddings, return prediction, Black–Litterman portfolio construction, and derivatives strategies. Section 4 reports the results, and Section 5 concludes.

## 2 Data

The HSI has expanded rapidly over the last decades, from an index dominated by financials and property to a broader, technology-heavy benchmark. To avoid survivorship bias, the investment universe on each date consists of the stocks that are HSI constituents on that date. A stock enters the universe on the effective date of its inclusion and leaves on the effective date of its removal, so no date uses a later list.

Daily open, high, low and close prices, adjusted closes and volumes come from Yahoo Finance for the stocks that were HSI members at some date between January 2005 and August 2026. We treat a volume of zero as missing and carry the previous volume forward, since most such records fall on half-day sessions, and a day with zero volume and an unchanged close is treated as a suspension. Where the provider's data are erroneous, namely one spin-off adjustment and nonpositive adjusted closes, the prices of the stock on that day are set to missing. Seven codes are dropped, four without Yahoo data, among them Hang Seng Bank (0011.HK), a constituent until January 2026, and three that Yahoo now assigns to another company. Market data include the HSI, the VHSI, the Tracker Fund of Hong Kong (2800.HK) as a proxy for the HSI TR, and the USD/HKD rate. The risk-free rate is approximated by the 1-month Hong Kong Interbank Offered Rate (HIBOR) fixing $R_t^f$ published by the Hong Kong Monetary Authority, and cash accrues on the actual/365 basis at the daily log return $r_t^f = \ln(1 + R_{t-1}^f \ell_t/365)$, with $\ell_t$ the number of calendar days since the previous trading day.

# 3 Methodology

At each monthly rebalance (Figure 1), the past year of each stock is embedded in a low-dimensional representation, a pooled regression maps the embedding to a forecast of the return over the next month, the Black–Litterman model turns the forecasts into weights, and index derivatives are added to the resulting stock portfolio.

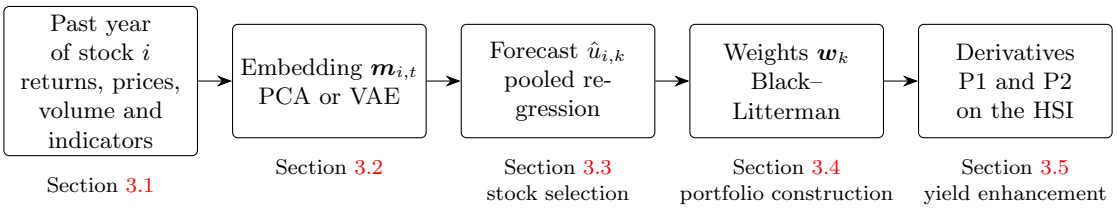



Figure 1. Monthly cycle of the strategy at each rebalance.

Let $P_{i,t}$ be the adjusted close of stock $i$ on trading day $t$ and $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$ its log return. The portfolio is rebalanced at the close of the first trading day of each month from September 2021 to August 2026, the dates $\tau_1, \ldots, \tau_{60}$, and every estimate used at $\tau_k$ relies on data up to the previous trading day $\tau_k - 1$.

## 3.1 Price path representation

The input of stock $i$ on day $t$ is the window $X_{i,t} \in \mathbb{R}^{L\times 16}$ of its $L = 240$ most recent trading days, the sequence length of Fischer and Krauss (2018). Each day includes 16 daily features adapted from Wu et al. (2026, Table 1), which are the return, open, high and low prices, volume and 11 technical indicators, transformed to be comparable across stocks, as listed in Table A.1. Both embeddings are fitted to describe the cumulative path of the standardized returns in the window,

$$c_{i,t,j} = \frac{1}{\sqrt{L}} \sum_{g=1}^{j} \frac{r_{i,t-L+g}}{s_{i,t}}, \qquad j = 1, \ldots, L, \tag{1}$$

where $s_{i,t}$ is the standard deviation of the returns in the window. The path $\boldsymbol{c}_{i,t} = (c_{i,t,1}, \ldots, c_{i,t,L})$ records when the gains and losses occurred, and its end value $c_{i,t,L}$ is the return over the window scaled by its volatility.

## 3.2 Dimension reduction by PCA and the VAE

This study examines two embeddings of dimension $d$, PCA and the VAE. PCA is linear and takes the return path alone as input, whereas the VAE is nonlinear and takes the whole window, so it can find a nonlinear low-dimensional manifold on which the paths lie and draw on the technical indicators. Both are refitted annually on the first trading day of each September from 2021 to 2025, on the windows before the refit except those of the final 252 trading days, which the VAE holds out for validation.

The linear embedding $\boldsymbol{m}_{i,t} \in \mathbb{R}^2$ consists of the scores of $\boldsymbol{c}_{i,t}$ on the first two principal components of the training paths. Two components account for 89.1% to 89.4% of the path variance at every refit, compared with 3.4% to 3.5% for the third, so the comparison uses $d = 2$ for both embeddings.

The VAE (Kingma and Welling, 2014) encodes the window $X_{i,t}$ with an LSTM into the mean $\boldsymbol{\mu}_{i,t}$ and the log variance $\ln \boldsymbol{\sigma}^2_{i,t}$ of a Gaussian approximate posterior over a latent variable $\boldsymbol{z} \in \mathbb{R}^d$,

and a second LSTM decodes a draw $\boldsymbol{z} = \boldsymbol{\mu}_{i,t} + \boldsymbol{\sigma}_{i,t} \odot \boldsymbol{\varepsilon}$, with $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, I)$, into a reconstruction $\hat{\boldsymbol{c}}_{i,t}$ of the path. The network minimizes the negative evidence lower bound for a Gaussian likelihood of variance $1/2$,

$$\mathcal{L}_{i,t} = \sum_{j=1}^{L} (c_{i,t,j} - \hat{c}_{i,t,j})^2 + \mathrm{KL}\left(N(\boldsymbol{\mu}_{i,t}, \operatorname{diag} \boldsymbol{\sigma}_{i,t}^2) \,\|\, N(\mathbf{0}, I)\right), \tag{2}$$

with KL the Kullback–Leibler divergence, and its embedding is the posterior mean $\boldsymbol{m}_{i,t} = \boldsymbol{\mu}_{i,t}$, with $d = 2$ or $d = 3$. All stocks share a single network, as in the pooled design of Fischer and Krauss (2018). Appendix A gives the architecture and the training.

### 3.3 Return prediction

Stock selection rests on a forecast of each stock's return relative to the others. The target is the excess return over the next month relative to the cross section, and a pooled regression on the embedding gives the forecast,

$$y_{i,t} = \sum_{j=1}^{21} \left(r_{i,t+j} - r^f_{t+j}\right), \qquad u_{i,t} = y_{i,t} - \bar{y}_t, \tag{3}$$

$$u_{i,t} = a + \boldsymbol{b}^\top \boldsymbol{m}_{i,t} + \nu_{i,t}, \qquad \hat{u}_{i,k} = \hat{a} + \hat{\boldsymbol{b}}^\top \boldsymbol{m}_{i,\tau_k - 1}, \tag{4}$$

where $\bar{y}_t$ is the mean of $y_{i,t}$ over the HSI members on day $t$, following the cross-sectional target of Fischer and Krauss (2018), $\nu_{i,t}$ is the error term, and $a$ and $\boldsymbol{b}$ are estimated at each refit by least squares pooled over the member days whose target ends before the refit. The forecast ranks the stocks, and its ranking power is measured by the rank IC, the Spearman correlation across stocks between the forecasts $\hat{u}_{i,k}$ and the returns from $\tau_k$ to $\tau_{k+1}$, and by the $t$-statistic of the mean IC over the 60 rebalances.

### 3.4 Portfolio construction

Portfolio construction turns the forecasts into weights with the Black–Litterman model. The universe $\mathcal{U}_k$ at $\tau_k$ is the set of $p_k$ HSI constituents with a year of complete returns and an embedding, and $p_k$ ranges from 57 to 93 over the backtest. The prior is the equilibrium return implied by the HSI as the market portfolio, and each view states that a stock beats its prior by its forecast relative to the universe mean $\bar{u}_k$. The uncertainty of the views is proportional to the prior covariance $\zeta \Sigma_k$, with $\zeta > 0$ a scale that cancels from the posterior (He and Litterman, 1999),

$$\begin{aligned} \Pi_{i,k} &= 21\, \delta \operatorname{Cov}\left(r_{i,t},\, r_{\mathrm{HSI},t}\right), \\ \boldsymbol{Q}_k &= \boldsymbol{\Pi}_k + \hat{\boldsymbol{u}}_k - \bar{u}_k \mathbf{1}, \\ \Omega_k &= \operatorname{diag}(\zeta \Sigma_k), \end{aligned} \tag{5}$$

where $\delta = 2.5$ is the risk-aversion coefficient of He and Litterman (1999), $r_{\mathrm{HSI},t}$ the daily log return of the HSI, Cov the sample covariance over the 252 trading days before $\tau_k$, $\hat{\boldsymbol{u}}_k$ the vector of the forecasts $\hat{u}_{i,k}$, and $\Sigma_k$ the monthly covariance matrix, estimated from the same 252 daily returns with the noise eigenvalues of the correlation matrix cleaned (Appendix B). With the identity matrix as the pick matrix, the posterior mean is

$$\boldsymbol{\mu}_k^{\mathrm{BL}} = \boldsymbol{\Pi}_k + \Sigma_k \left(\Sigma_k + \operatorname{diag} \Sigma_k\right)^{-1} \left(\hat{\boldsymbol{u}}_k - \bar{u}_k \mathbf{1}\right). \tag{6}$$

Each stock portfolio maximizes the mean-variance objective under a long-only constraint and a single-name cap of 3%,

$$\boldsymbol{w}_k = \arg\max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \boldsymbol{\mu}_k - \frac{\delta}{2} \boldsymbol{w}^\top \Sigma_k \boldsymbol{w} \quad \text{subject to} \quad \mathbf{1}^\top \boldsymbol{w} = 1, \quad 0 \le w_i \le 0.03. \tag{7}$$

An equally weighted portfolio (EW) serves as a naive benchmark, and the other portfolios differ only in the expected returns $\boldsymbol{\mu}_k$ (Table 1). M1 uses the prior alone, so M1 against the HSI TR measures portfolio construction. M2 adds the views, so M2 against M1 measures stock selection. Every portfolio is rebalanced monthly and pays 15 basis points one way on turnover, and a suspended stock keeps its weight.

| Portfolio | Expected returns $\boldsymbol{\mu}_k$ | Role |
|---|---|---|
| EW | none, $w_i = 1/p_k$ | naive benchmark |
| M1 | $\boldsymbol{\Pi}_k$ | portfolio construction without stock selection |
| M2-PCA | $\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the PCA forecast | stock selection with the linear embedding |
| M2-VAE | $\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the VAE forecast, $d = 2$ | stock selection with the nonlinear embedding |
| M3 | $\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the VAE forecast, $d = 3$ | stock selection with a larger nonlinear embedding |

Table 1. Stock portfolios. All except EW solve Equation (7) with the stated expected returns.

### 3.5 Derivatives strategies

The derivatives are written on the HSI, the price index of the benchmark. Each position holds $n = \left\lfloor \frac{h\,\beta_k\,E_t}{50\,S_t} \right\rfloor$ contracts, where $E_t$ is the value of the stock portfolio on the entry day $t$, $\beta_k$ its ex-ante beta to the HSI over the 252 days before $\tau_k$, $S_t$ the HSI close, 50 the contract multiplier in HKD and $h$ the coverage.

We set the base case to $h = 0.5$, which covers 49% of the stock portfolio on average, and report $h = 1$ in Appendix E. Positions are opened in the front-month contract, at the first rebalance and at each expiry. Options are priced by the model of Black and Scholes (1973) with a continuous dividend yield (Merton, 1973) at the VHSI level $v_t$, and every trade pays transaction costs. Appendix C sets out the contract terms, the costs and the cash held for margin. Four scenarios are compared: no derivatives (None), the yield strategy P1, the protection strategy P2, and P1 and P2.

P1 targets the volatility risk premium. Implied variance tends to exceed the variance later realized (Carr and Wu, 2009), and delta-hedged long option positions lose on average (Bakshi and Kapadia, 2003), a premium their sellers earn. A covered call collects this premium but also sells equity exposure and holds an uncompensated timing position (Israelov and Nielsen, 2015), which lowers the expected return of a portfolio that aims to beat the index. P1 therefore sells $n$ out-of-the-money (OTM) calls at the first listed strike at or above $1.04 S_t$ and hedges them with $H_t = \mathrm{round}\left( \frac{-\Delta_t}{e^{(R_t^f - q)\vartheta_t}} \right)$ HSI futures reset at every close, where $\Delta_t$ is the delta of the short calls in index units, $\vartheta_t$ the time to expiry and $q$ the dividend yield. At a constant volatility, the hedged position gains

$$G \approx \frac{50}{2}\, n \sum_t \Gamma_t S_t^2 \left( \frac{(v_0/100)^2}{A} - r_{\mathrm{HSI},t+1}^2 \right) \tag{8}$$

in HKD, with $t$ running from the sale to expiry, $\Gamma_t$ the gamma of one call, $v_0$ the VHSI at which the calls are sold and $A$ the number of trading days per year. P1 therefore gains when the index moves less than the VHSI implied.

P2 limits losses in falling markets at a low cost. Puts are costly protection, with strongly negative expected returns (Coval and Shumway, 2001). P2 therefore holds a put spread collar only when the VHSI signals stress, in the spirit of the adjusted collars of Szado and Schneeweis (2010), and finances the protection within the position.

The strategy triggers when the VHSI closes above 30 and exits when it closes below 22. Each position buys $n$ puts at the first listed strike $K_1$ at or below $0.95S_t$, sells $n$ puts at the first listed strike $K_2$ at or below $0.90S_t$, and sells $n$ calls at the first listed strike $K_3$ at or above the strike within $[1.05S_t, 1.10S_t]$ that sets the net premium to zero. Figure 2 shows the resulting payoff. The collar absorbs losses between $K_1$ and $K_2$ and gives up gains above $K_3$, and selling the lower put sells the option with the lower expected return (Coval and Shumway, 2001). The size is capped at entry so that the absolute delta does not exceed 15% of the net asset value, a cap that binds at 12 of the 19 entries at $h = 0.5$.

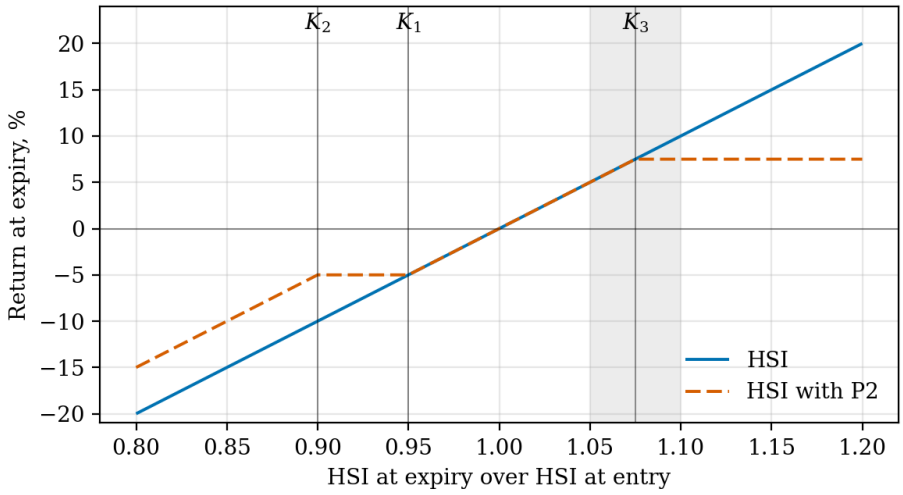



Figure 2. Return at expiry of the HSI and of the HSI with the P2 collar.

To isolate the design choices, we examine two variants. The first, P1 unhedged, holds the P1 calls without futures, which is a covered call. The second, P2, full trigger, also turns on when the stock portfolio falls more than 5% below its running peak, and turns off only when the VHSI is below 22 and the drawdown is within 2%.

# 4 Results

Section 4.1 compares the forecasting performance of PCA and the VAE, Section 4.2 examines whether the forecast transfers into active return, Section 4.3 evaluates the derivatives strategies, and Section 4.4 decomposes the outperformance of the final portfolio by source. Performance is measured by annualized return, volatility, Sharpe ratio and maximum drawdown (MDD) and, against a benchmark, by active return, tracking error (TE) and IR, as defined in Appendix D.

## 4.1 Predictive information in the embeddings

Both embeddings carry predictive information about next-month returns (Table 2). The forecasts from PCA and from the VAE with $d = 2$ have mean ICs of 0.072 and 0.078, with $t$-statistics of 2.47 and 2.66. The VAE with $d = 3$ is weaker and not significant. The pooled regression explains

| | Mean IC by year | | | | | All months | | |
|---|---|---|---|---|---|---|---|---|
| Embedding | 2021/22 | 2022/23 | 2023/24 | 2024/25 | 2025/26 | Mean | $t$ | IC $> 0$, % |
| PCA | 0.094 | 0.054 | 0.136 | 0.035 | 0.040 | 0.072 | 2.47 | 63 |
| VAE ($d = 2$) | 0.098 | 0.069 | 0.138 | 0.039 | 0.047 | 0.078 | 2.66 | 63 |
| VAE ($d = 3$) | 0.084 | 0.050 | 0.026 | 0.024 | 0.040 | 0.045 | 1.49 | 58 |

Table 2. Rank IC of each forecast, averaged within each year from September to August and over all 60 months, with the $t$-statistic of the mean, which treats the months as independent, and the share of months with a positive IC.

at most 0.13% of the variance of monthly relative returns, so the forecast is informative about the ranking of stocks rather than the size of their relative returns.

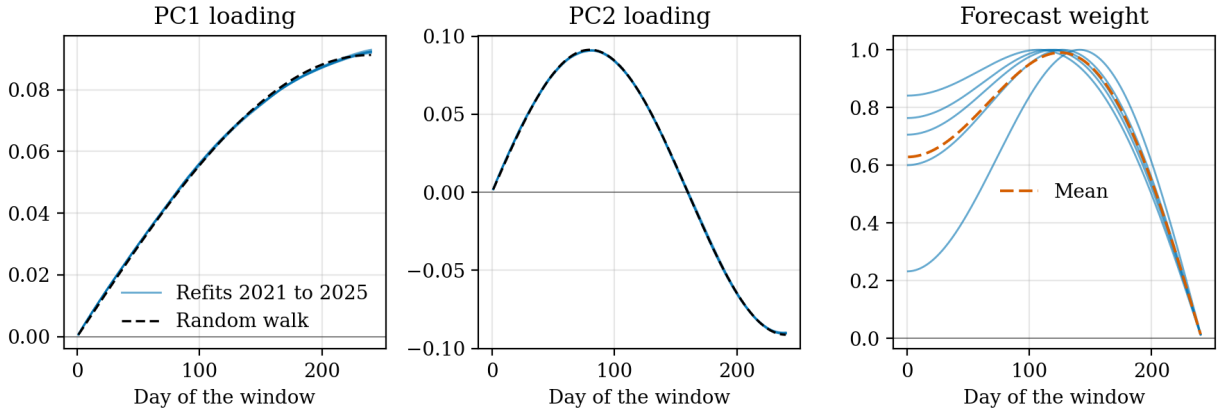



Figure 3. Loadings of the first two principal components at each refit from 2021 to 2025 with the eigenvectors of a random walk, and the forecast weight $\eta_g$ relative to the largest weight.

The loadings of the linear embedding reveal that the standardized return path of an HSI stock has the covariance structure of a random walk (Figure 3). At every refit, the loadings of the first two principal components coincide with the eigenvectors of the covariance matrix of a random walk, and their shares of the variance, about 80% and 9%, match the 81% and 9% of a random walk. The first component weights the whole window and tracks the volatility-scaled return over the year, whereas the second contrasts the first two-thirds of the window with the last third and so measures whether the gains came early or late.

Because the path is cumulative, the forecast is a weighted sum of the standardized daily returns,

$$\hat{u}_{i,k} = \text{const} + \sum_{g=1}^{L} \eta_g \frac{r_{i,\tau_k-1-L+g}}{s_{i,\tau_k-1}}, \qquad \eta_g = \frac{1}{\sqrt{L}} \sum_{j=g}^{L} \left( \hat{b}_1 \chi_{1,j} + \hat{b}_2 \chi_{2,j} \right), \tag{9}$$

where $\chi_{1,j}$ and $\chi_{2,j}$ are the loadings of the two components and $\hat{b}_1$ and $\hat{b}_2$ their coefficients. As shown in the right panel of Figure 3, the forecast weights are largest for returns from 3 to 9 months before the forecast and small for the most recent month, so the forecast reflects momentum over the past year (Jegadeesh and Titman, 1993).

Random-walk behavior also explains why the nonlinear embedding does not outperform the linear one. If the standardized returns are also independent and identically distributed, the path

is approximately a Brownian motion by Donsker's theorem, a Gaussian process that its covariance describes fully, so a nonlinear embedding has little to find beyond the directions of largest variance, and a VAE with linear maps recovers exactly the principal subspace (Lucas et al., 2019). The technical indicators do not alter this, because the VAE is trained to reconstruct the return path and most indicators are functions of the same closing prices. Consistent with this, the PCA scores explain the VAE embedding with an $R^2$ above 0.9 in four of the five refits (Table A.2). The monthly ICs of the two forecasts have a correlation of 0.99, although the IC of the VAE is marginally higher in every year, and the mean IC difference of 0.006 has a paired $t$-statistic of 1.23. With $d = 3$, one dimension collapses to the prior in four refits and adds no information, and the IC falls most in 2023/24, when all three dimensions are active (Tables 2 and A.2).

### 4.2 Stock portfolios

| Portfolio | Return | Volatility | Sharpe ratio | Max drawdown | Turnover | Avg. # holdings |
|---|---|---|---|---|---|---|
| EW | 2.0 | 24.1 | −0.04 | −39.8 | 8.0 | 78 |
| M1 | 2.2 | 25.3 | −0.03 | −43.0 | 18.6 | 42 |
| M2-PCA | 5.3 | 23.9 | 0.09 | −40.6 | 20.4 | 40 |
| M2-VAE | 4.7 | 24.0 | 0.07 | −40.6 | 21.8 | 40 |
| M3 | 3.8 | 24.0 | 0.03 | −40.7 | 25.6 | 40 |
| HSI TR | 3.1 | 24.6 | 0.00 | −41.7 | | |

Table 3. Performance of the stock portfolios from September 2021 to August 2026. Return, volatility and maximum drawdown are in %, the first two annualized, and turnover is the mean sum of absolute weight changes per rebalance, in %.

In the backtest, stock selection turns the forecast into outperformance, whereas portfolio construction alone does not. EW and M1 underperform the HSI TR, while M2-PCA and M2-VAE outperform it with lower volatility (Table 3).

M2-PCA beats M1 in every year, and M2-VAE beats it in four of the five years (Table 5). Against M1, which isolates stock selection, M2-PCA and M2-VAE earn active returns of 2.9% and 2.4% a year, with $t$-statistics of 1.70 and 1.40, short of the 5% critical value of 1.96 (Table 4). Their outperformance over the HSI TR accrues in four of the five years rather than in a single market environment, and it is 0.0% for M2-PCA and −1.3% for M2-VAE in 2024/25, when the HSI TR rose 43.2% (Table 5 and Figures 4 and 5). With a beta of 0.95 to the HSI TR, the M2 portfolios do not owe their active return to market exposure.

The difference between M2-PCA and M2-VAE, 0.57% a year with a $t$-statistic of 1.24, is not significant. The two portfolios share 95% of their weights, as expected from forecasts that rank stocks alike. M3 inherits the weaker forecast of its VAE and earns less than the M2 portfolios.

### 4.3 Derivatives strategies

The derivatives strategies have the same effect on all stock portfolios to within 0.07% a year, so the discussion refers to M2-PCA (Table 6).

Yield enhancement requires the delta hedge. Without futures, the covered call (P1 unhedged) lowers volatility by selling index exposure but also reduces returns, whereas P1 raises the return by 0.73% a year at unchanged volatility, an active return of 0.69% with a $t$-statistic of 1.43. This finding is consistent with a volatility risk premium in HSI options, as the VHSI exceeded the subsequently realized volatility in 44 of the 60 months.

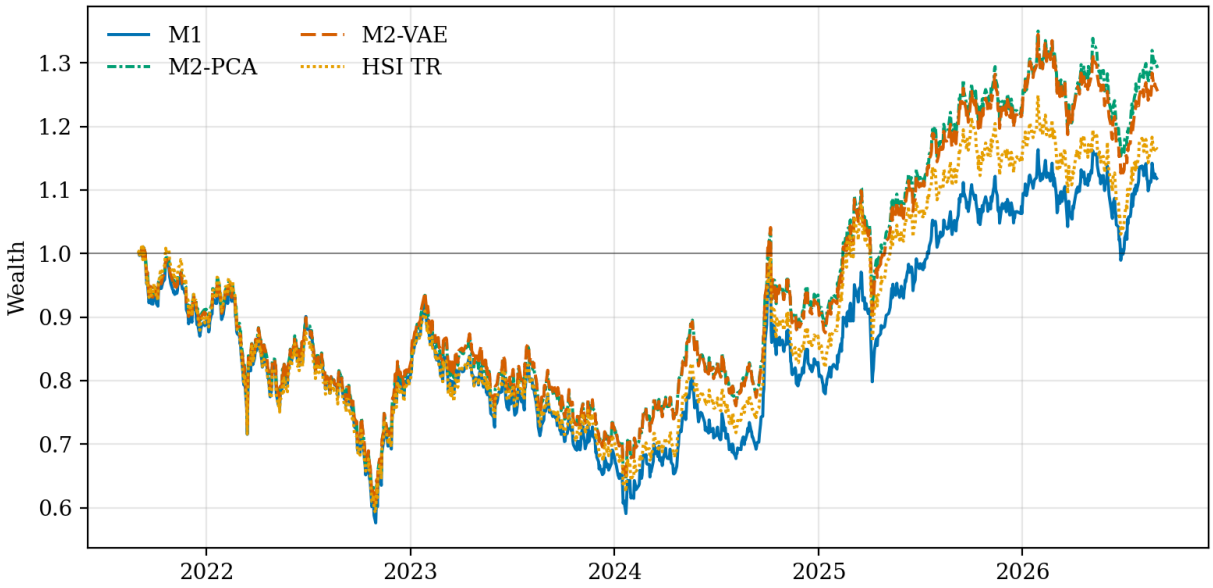



Figure 4. Wealth of M1, M2-PCA, M2-VAE and the HSI TR.

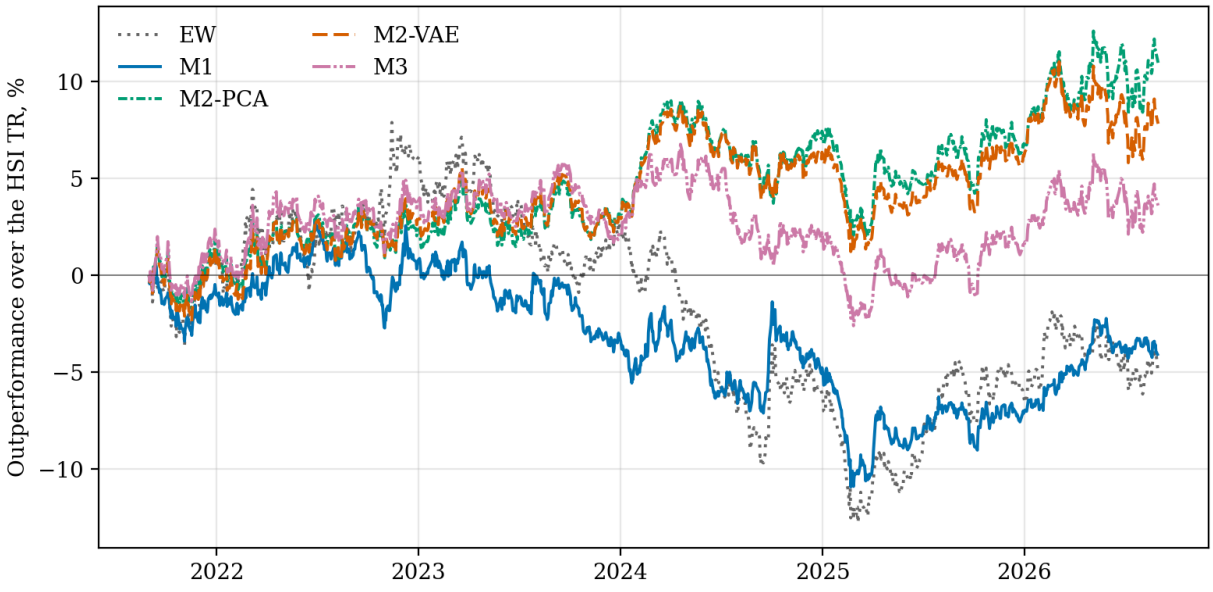



Figure 5. Outperformance of each stock portfolio over the HSI TR, in %.

| | Against the HSI TR | | | | | Against M1 | | |
|---|---|---|---|---|---|---|---|---|
| Portfolio | Beta | Active | TE | IR | $t$ | Active | IR | $t$ |
| EW | 0.94 | −1.0 | 6.7 | −0.16 | −0.35 | | | |
| M1 | 1.01 | −0.8 | 5.2 | −0.16 | −0.36 | | | |
| M2-PCA | 0.95 | +2.1 | 5.2 | +0.40 | +0.89 | +2.9 | +0.76 | +1.70 |
| M2-VAE | 0.95 | +1.5 | 5.2 | +0.29 | +0.65 | +2.4 | +0.62 | +1.40 |
| M3 | 0.95 | +0.7 | 5.2 | +0.14 | +0.30 | +1.6 | +0.39 | +0.88 |

Table 4. Active performance of the stock portfolios against the HSI TR and against M1. Active return and TE are in % a year.

| Portfolio | 2021/22 | 2022/23 | 2023/24 | 2024/25 | 2025/26 |
|---|---|---|---|---|---|
| EW | +1.3 | −0.6 | −9.1 | +5.2 | −0.6 |
| M1 | +0.9 | −2.1 | −4.6 | −2.1 | +3.1 |
| M2-PCA | +1.6 | +1.3 | +3.1 | 0.0 | +4.3 |
| M2-VAE | +1.4 | +1.9 | +2.5 | −1.3 | +2.4 |
| M3 | +2.3 | +1.5 | −1.5 | −2.4 | +2.4 |
| HSI TR | −21.0 | −3.8 | 2.3 | 43.2 | 4.7 |

Table 5. Outperformance of the stock portfolios over the HSI TR in each September-to-August year, in %, with the HSI TR return in the last row.

| | | Performance | | | | Over none | | Over HSI TR | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Portfolio | Scenario | Return | Vol. | Sharpe | MDD | Gain | $t$ | Active | IR | Days |
| M2-PCA | None | 5.24 | 22.8 | 0.10 | −39.2 | | | +2.06 | 0.38 | |
| | P1 | 5.97 | 22.9 | 0.13 | −38.6 | +0.73 | 1.43 | +2.75 | 0.51 | |
| | P2 | 5.55 | 21.1 | 0.12 | −35.5 | +0.31 | 0.21 | +2.36 | 0.36 | 31 |
| | P1 and P2 | 6.28 | 21.2 | 0.15 | −34.9 | +1.04 | 0.68 | +3.05 | 0.46 | 31 |
| | P1 unhedged | 5.02 | 19.9 | 0.10 | −34.3 | −0.22 | −0.10 | +1.85 | 0.23 | |
| | P2, full trigger | 3.24 | 19.7 | 0.01 | −35.8 | −2.00 | −1.04 | +0.15 | 0.02 | 85 |
| M2-VAE | None | 4.66 | 22.9 | 0.07 | −39.2 | | | +1.52 | 0.28 | |
| | P1 | 5.39 | 23.0 | 0.10 | −38.6 | +0.72 | 1.42 | +2.20 | 0.41 | |
| | P2 | 4.97 | 21.2 | 0.09 | −35.5 | +0.31 | 0.21 | +1.81 | 0.28 | 31 |
| | P1 and P2 | 5.70 | 21.3 | 0.12 | −34.9 | +1.04 | 0.68 | +2.50 | 0.38 | 31 |
| | P1 unhedged | 4.45 | 20.0 | 0.07 | −34.2 | −0.22 | −0.10 | +1.31 | 0.17 | |
| | P2, full trigger | 2.64 | 19.8 | −0.02 | −35.8 | −2.02 | −1.05 | −0.43 | −0.06 | 86 |

Table 6. Derivatives strategies on M2-PCA and M2-VAE at coverage $h = 0.5$. Gain is the difference in annual return from no derivatives and $t$ the $t$-statistic of the active return over no derivatives, active return and IR are measured against the HSI TR, and days is the share of days on which P2 is held, in %.

Conditioning the collar on the VHSI keeps protection costs low. P2 is held on 31% of days, raises the return by 0.31% a year and lowers the MDD from 39.2% to 35.5%, whereas P2, full trigger, holds the collar on 85% of days and lowers the return by 2.00% a year. The gain of P2 arises in the decline of 2022 (Figure 6). The collar was held from March 2022 to April 2023, and its short call gave up the whole gain in the rebound from November 2022 before the collar closed with a gain in April 2023. The three later spells of P2, in 2023 to 2025, left the relative value almost unchanged.

Together, P1 and P2 raise the return by 1.04% a year, an active return of 0.98% over no derivatives, and reduce both the volatility and the MDD.

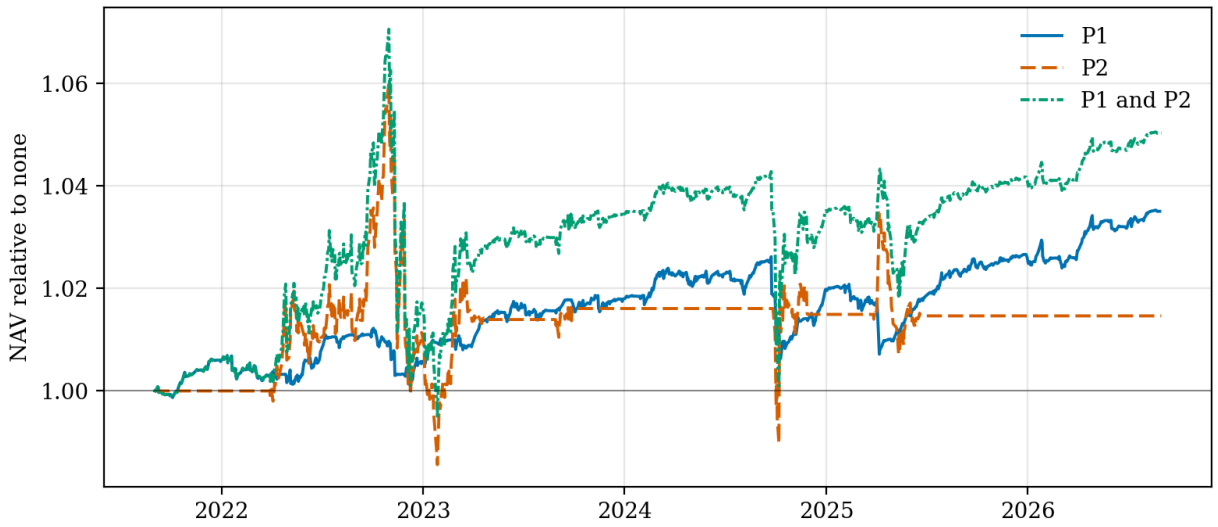



Figure 6. Net asset value of M2-PCA with derivatives strategies relative to that without derivatives.

### 4.4 Sources of outperformance

| | | M2-PCA | | | M2-VAE | | |
|---|---|---|---|---|---|---|---|
| Source | Comparison | Active | IR | $t$ | Active | IR | $t$ |
| Portfolio construction | M1 against the HSI TR | −0.85 | −0.16 | −0.36 | −0.85 | −0.16 | −0.36 |
| Stock selection | M2 against M1 | +2.93 | +0.76 | +1.70 | +2.35 | +0.62 | +1.40 |
| Derivatives, P1 | P1 against none | +0.69 | +0.64 | +1.43 | +0.69 | +0.64 | +1.42 |
| Derivatives, P2 | P1 and P2 against P1 | +0.29 | +0.09 | +0.21 | +0.30 | +0.10 | +0.21 |
| Total | P1 and P2 against the HSI TR | +3.05 | +0.46 | +1.03 | +2.50 | +0.38 | +0.86 |

Table 7. Sources of the active return of M2-PCA and M2-VAE with P1 and P2 against the HSI TR at $h = 0.5$. Each step is measured against the previous one, and the steps add up to the total except for the cash held for margin, which changes the active return by −0.02% for M2-PCA and +0.01% for M2-VAE. None is the portfolio without derivatives. Active returns are differences of annualized log returns, so they differ slightly from the differences of the returns in Table 6.

Stock selection is the main source of the outperformance of the final portfolio, M2-PCA with P1 and P2 (Table 7). Portfolio construction alone loses 0.85% a year against the HSI TR, stock

selection adds 2.93%, and P1 and P2 add 0.69% and 0.29%, for an active return of 3.05% with an IR of 0.46. None of these steps is significant at the 5% level, the largest $t$-statistic being 1.70 for stock selection, whereas the mean IC of the same forecast has a $t$-statistic of 2.47 (Table 2). M2-VAE follows the same pattern, with stock selection adding 2.35%.

## 5 Conclusion

In this study, we examine (i) whether embeddings of the past year of returns and technical indicators of HSI stocks carry information about their returns over the next month, (ii) whether a nonlinear embedding improves on a linear one, and (iii) what the embeddings deliver in a long-only portfolio with index derivatives. We embed each stock's return path with PCA and its whole trading history with an LSTM VAE, forecast returns with a pooled regression on the embedding, construct weights with the Black–Litterman model, and enhance yield with delta-hedged short calls and a put spread collar held when the VHSI is high.

The embeddings carry significant information about returns, and the forecast they produce reflects momentum over the past year. The VAE and PCA perform similarly, because the standardized return path behaves like a random walk that leaves little structure beyond its principal components. From September 2021 to August 2026, the PCA portfolio with derivatives strategies outperforms the HSI TR by 3.05% a year with lower volatility and drawdown. The outperformance mainly stems from stock selection, while the hedged calls enhance yield and the conditional collar protects at little cost. However, neither the transfer of the forecast into active return nor the gain from the derivatives is statistically significant.

Pricing at the VHSI is a limitation of this study. The VHSI is computed from HSI options across strikes using the VIX method (Hang Seng Indexes Company Limited, 2023), so its square approximates a 30-day variance swap rate (Carr and Wu, 2009) rather than the implied volatility of any traded strike. Under a negative skew, OTM calls trade at implied volatilities below the VHSI and OTM puts above it. The backtest gain from the derivatives disappears if calls are priced about 1.6 volatility points below the VHSI and puts about 1.6 points above it.

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# A The input features and the VAE

Table A.1 lists the input features, Figure A.1 draws the architecture of the VAE, and Table A.2 reports the diagnostics of its embeddings at each refit.

The encoder and decoder have two LSTM layers with 32 hidden units and dropout 0.25. Training uses Adam with a learning rate of 0.001, batches of 256 windows and gradient norm clipping at 1, and runs for at most 300 epochs with early stopping after 10 epochs without improvement in the validation loss.

| Feature | Definition |
|---|---|
| Open | $P^{\mathrm{o}}_t / P^{\mathrm{c}}_{t-1} - 1$ |
| High | $P^{\mathrm{h}}_t / P^{\mathrm{c}}_{t-1} - 1$ |
| Low | $P^{\mathrm{l}}_t / P^{\mathrm{c}}_{t-1} - 1$ |
| Return | $r_t = \ln(P_t / P_{t-1})$ |
| Volume | $\ln \mathcal{V}_t$ |
| True range | $\mathrm{TR}_t / P^{\mathrm{c}}_t$, with $\mathrm{TR}_t = \max\left(P^{\mathrm{h}}_t - P^{\mathrm{l}}_t, \lvert P^{\mathrm{h}}_t - P^{\mathrm{c}}_{t-1}\rvert, \lvert P^{\mathrm{l}}_t - P^{\mathrm{c}}_{t-1}\rvert\right)$ |
| Average true range | mean of $\mathrm{TR}_t$ over 20 days, over $P^{\mathrm{c}}_t$ |
| Simple moving averages | $P^{\mathrm{c}}_t$ over its mean over 5, 10 or 20 days, less one |
| Exponential moving averages | $P^{\mathrm{c}}_t$ over its exponential average with span 5, 10 or 20 days, less one |
| Stochastic %K | $(P^{\mathrm{c}}_t - \min P^{\mathrm{l}}) / (\max P^{\mathrm{h}} - \min P^{\mathrm{l}})$ over 14 days |
| Stochastic %D | mean of %K over 3 days |
| Relative strength index | average rise of $P^{\mathrm{c}}$ over its average rise plus average fall, smoothing 1/14 |

Table A.1. The 16 daily input features of the VAE on day $t$, adapted from Wu et al. (2026, Table 1). $P^{\mathrm{o}}$, $P^{\mathrm{h}}$, $P^{\mathrm{l}}$ and $P^{\mathrm{c}}$ are the split-adjusted open, high, low and close, $P$ is the dividend- and split-adjusted close, and $\mathcal{V}$ the volume. Within each window of $L$ days, the return is divided by its standard deviation and the log volume is demeaned, and the other features are standardized with the means and standard deviations over the training days.

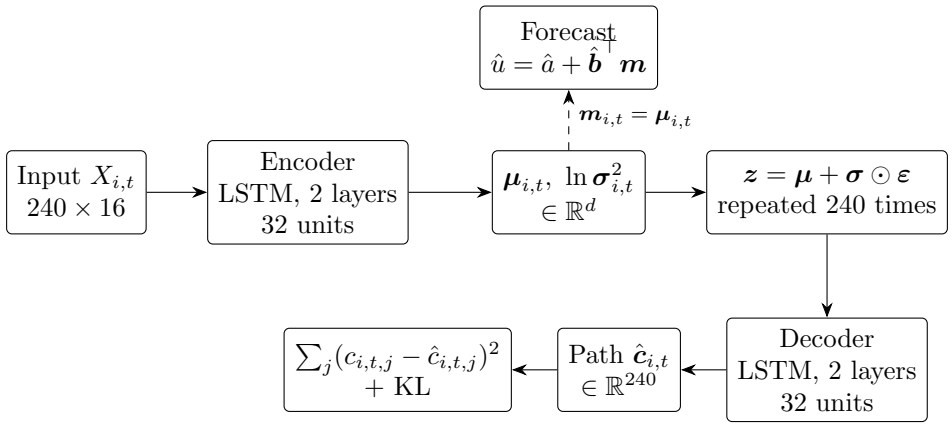



Figure A.1. Architecture of the VAE. The encoder reads 240 days of 16 features and returns the mean and log variance of the latent variable, and the decoder receives a draw of the latent variable at each of 240 steps and reconstructs the cumulative path of standardized returns. The posterior mean $\boldsymbol{m}_{i,t}$ is the embedding that enters the forecast.

| | | Active units | | $R^2$ on PCA scores | |
|---|---|---|---|---|---|
| Refit | Training windows | $d = 2$ | $d = 3$ | Dimension 1 | Dimension 2 |
| 2021 | 154,401 | 2 | 2 | 0.91 | 0.97 |
| 2022 | 167,233 | 2 | 2 | 0.91 | 0.97 |
| 2023 | 182,816 | 2 | 3 | 0.94 | 0.97 |
| 2024 | 201,166 | 2 | 2 | 0.68 | 0.91 |
| 2025 | 220,792 | 2 | 2 | 0.92 | 0.96 |

Table A.2. Diagnostics of the VAE embeddings at each refit. Active units count the latent dimensions whose posterior mean has a variance across windows above 0.01 (Burda et al., 2016). $R^2$ is that of each dimension of the embedding of the VAE with $d = 2$ regressed on the two PCA scores of the same windows.

## B Covariance estimation

The covariance matrix at $\tau_k$ is estimated from the 252 daily returns before $\tau_k$ of the stocks in $\mathcal{U}_k$. Let $D_k$ be the diagonal matrix of their variances and $C_k = \Xi_k \Lambda_k \Xi_k^\top$ their correlation matrix, with eigenvectors $\Xi_k$ and eigenvalues $\Lambda_k = \mathrm{diag}(\lambda_1, \ldots, \lambda_{p_k})$ in decreasing order. The eigenvalues above the upper edge of the Marchenko–Pastur law (Marchenko and Pastur, 1967) are kept as factors, at least one and at most four, and the others are replaced by their mean $\bar{\lambda}_k$ (Laloux et al., 2000),

$$f_k = \min\left(\max\left(\#\left\{\lambda \in \{\lambda_1, \ldots, \lambda_{p_k}\} : \lambda > \left(1 + \sqrt{p_k/252}\right)^2\right\}, 1\right), 4\right), \tag{10}$$

$$\tilde{\Lambda}_k = \mathrm{diag}\left(\lambda_1, \ldots, \lambda_{f_k}, \bar{\lambda}_k, \ldots, \bar{\lambda}_k\right). \tag{11}$$

The monthly covariance matrix is

$$\Sigma_k = 21\, D_k^{1/2}\, \tilde{C}_k\, D_k^{1/2}, \tag{12}$$

where $\tilde{C}_k$ is $\Xi_k \tilde{\Lambda}_k \Xi_k^\top$ rescaled to a unit diagonal.

## C Derivatives contracts and costs

The portfolio starts with USD 100 million, converted to HKD, and holds stocks worth $E_t$, cash $B_t$ earning $r_t^f$ and derivatives worth $O_t$, so that its net asset value is $V_t = E_t + B_t + O_t$. At each rebalance, the cash share is set in proportion to the initial margin of the futures,

$$\phi_k = \rho\, h\, \beta_k, \qquad \rho = \frac{117{,}705}{50\, S_{\mathrm{M}}}, \tag{13}$$

where $\rho$ is the initial margin of HKD 117,705 for one HSI futures contract as a share of its notional value at the index level $S_{\mathrm{M}}$ of 2 March 2026. The stock portfolio is then reset to $E_{\tau_k} = (1 - \phi_k) V_{\tau_k}$ at a cost of 15 basis points. Every scenario holds the same cash share, so the scenarios differ only by their derivatives.

HSI options and futures have a multiplier of HKD 50 and expire on the second-to-last trading day of the month, and listed strikes are 100 points apart below 20,000 and 200 points apart above. Options are valued at the VHSI with the HIBOR fixing and a dividend yield $q$ of 3%, the rounded average trailing yield of 2800.HK from 2014 to August 2021, and are cash settled at the closing

level on expiry at no cost. Each option trade pays half of the widest spread the exchange permits its market makers to quote,

$$\psi(x) = \begin{cases} \max(30,\, 0.1x) & x \le 750, \\ 75 & x > 750, \end{cases} \tag{14}$$

in index points for a premium of $x$ points, and each option or futures contract pays an exchange fee and levy of HKD 10.54. The futures price $F_t = S_t e^{(R_t^f - q)\vartheta_t}$ is marked to market daily, and each futures trade also pays a half spread of 1 index point, the minimum tick.

# D Performance measures

Let $r_t^{\mathrm{p}}$ and $r_t^{\mathrm{b}}$ be the daily log returns of a portfolio and of a benchmark, $t = 1, \ldots, T$, over a sample of $Y$ years with $A$ trading days per calendar year. For a daily series $x_t$, write

$$s(x) = \{(T-1)^{-1} \sum_t (x_t - \bar{x})^2\}^{1/2}, \qquad \bar{x} = T^{-1} \sum_t x_t. \tag{15}$$

With the excess returns $e_t^{\mathrm{p}} = r_t^{\mathrm{p}} - r_t^f$ and $e_t^{\mathrm{b}} = r_t^{\mathrm{b}} - r_t^f$, the daily active return $\tilde{r}_t$ and the wealth $W_t$,

$$\tilde{r}_t = r_t^{\mathrm{p}} - r_t^{\mathrm{b}}, \qquad W_t = \exp(\sum_{s \le t} r_s^{\mathrm{p}}), \tag{16}$$

$$\begin{aligned}
\text{Return} &= e^{A\bar{r}^{\mathrm{p}}} - 1, & \text{Vol} &= \sqrt{A}\, s(r^{\mathrm{p}}), \\
\text{SR} &= \sqrt{A}\, \frac{\bar{e}^{\mathrm{p}}}{s(e^{\mathrm{p}})}, & \text{MDD} &= \min_t \left( \frac{W_t}{\max\left(1, \max_{s \le t} W_s\right)} - 1 \right), \\
\text{Active} &= A\, \bar{\tilde{r}}, & \text{TE} &= \sqrt{A}\, s(\tilde{r}), \\
\text{IR} &= \frac{\text{Active}}{\text{TE}}, & t_{\text{IR}} &= \text{IR}\sqrt{Y}, \\
\beta^{\mathrm{p}} &= \frac{\sum_t \left(e_t^{\mathrm{p}} - \bar{e}^{\mathrm{p}}\right)\left(e_t^{\mathrm{b}} - \bar{e}^{\mathrm{b}}\right)}{\sum_t \left(e_t^{\mathrm{b}} - \bar{e}^{\mathrm{b}}\right)^2},
\end{aligned}$$

where Return, Vol and SR are the annualized return, volatility and Sharpe ratio, Active is the active return, a difference of annualized log returns, $t_{\text{IR}}$ its $t$-statistic and $\beta^{\mathrm{p}}$ the beta. A two-sided test of a zero mean at the 5% level against the standard normal distribution requires $|t_{\text{IR}}| \ge 1.96$, that is $|\text{IR}| \ge 1.96/\sqrt{Y} = 0.88$ for the $Y = 4.99$ years of the backtest.

# E Derivatives strategies at full coverage

Doubling the coverage to $h = 1$ nearly doubles the gain of P1 and leaves that of P2 almost unchanged (Table E.1). On M2-PCA, P1 adds 1.38% a year, P2 adds 0.34% and the two strategies together add 1.72%, for an active return of 3.66% against the HSI TR with an IR of 0.50. P2 changes little because the delta cap binds at all 19 of its entries.

| | | Performance | | | | Over none | | Over HSI TR | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Portfolio | Scenario | Return | Vol. | Sharpe | MDD | Gain | $t$ | Active | IR | Days |
| M2-PCA | None | 5.21 | 21.8 | 0.10 | −37.7 | | | +2.04 | 0.36 | |
| | P1 | 6.60 | 22.0 | 0.16 | −36.6 | +1.38 | 1.42 | +3.35 | 0.57 | |
| | P2 | 5.55 | 20.0 | 0.12 | −33.9 | +0.34 | 0.23 | +2.36 | 0.33 | 31 |
| | P1 and P2 | 6.94 | 20.2 | 0.19 | −32.7 | +1.72 | 0.99 | +3.66 | 0.50 | 31 |
| M2-VAE | None | 4.66 | 21.9 | 0.07 | −37.7 | | | +1.51 | 0.27 | |
| | P1 | 6.05 | 22.1 | 0.13 | −36.5 | +1.39 | 1.42 | +2.83 | 0.49 | |
| | P2 | 5.00 | 20.1 | 0.10 | −33.9 | +0.34 | 0.23 | +1.84 | 0.26 | 31 |
| | P1 and P2 | 6.39 | 20.3 | 0.16 | −32.6 | +1.73 | 0.99 | +3.15 | 0.44 | 31 |

Table E.1. Derivatives strategies on M2-PCA and M2-VAE at coverage $h = 1$, with columns as in Table 6.